

QUESTION 1 (12 marks)

(a) Differentiate i)  $y = 3x^2 - \frac{1}{x}$  [2]

ii)  $f(x) = (2 - 4x)^{\frac{1}{2}}$  [2]

iii)  $y = \frac{1-x}{x^2+1}$  [2]

(b) If  $\frac{dy}{dx} = 12x^2 + 1$  and  $y = 10$  when  $x = 1$ , find  $y$  when  $x = 2$  [3]

(c) Find the value of  $m$  if  $\int_m^{m+2} (2x) dx = 16$  [3]

QUESTION 2 (12 marks)      START A NEW PAGE

(a) For the curve  $y = 2x^3 - 3x^2 - 36x + 2$

i.) find the coordinates of any stationary points and determine their nature [3]

ii.) find any points of inflexion [2]

iii.) sketch the curve [2]

iv.) state the domain for which this curve is decreasing [1]

(b) If  $\alpha$  and  $\beta$  are the roots of the equation  $2x^2 - 8x - 25 = 0$  [4]

find i)  $\alpha + \beta$

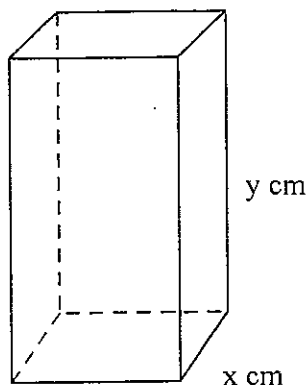
ii)  $\alpha\beta$

iii)  $\alpha^2 + \beta^2$

**QUESTION 3 (12 marks) START A NEW PAGE**

(a) A box, open at the top, is to be made from cardboard.

The base of the box is a square of side  $x$  cm and its height is  $y$  cm, as shown in the diagram.



i) If the volume of the box is  $32 \text{ cm}^3$ , [1]

show that  $y = \frac{32}{x^2}$

ii) Show that the area of cardboard needed will be [1]

given by  $A = x^2 + \frac{128}{x} \text{ cm}^2$

iii) Find the dimensions of the box if this area is to be a minimum. [4]

(b) A parabola **T** has equation  $x^2 = 4(2 - y)$ .

i) Write this equation in the form  $(x - h)^2 = 4a(y - k)$  and hence find the coordinates of its vertex and focus [2]

ii) Draw a sketch of **T**, clearly indicating the coordinates of the vertex and focus [1]

iii) Find the equation of the normal to the parabola **T** at the point (2,1) [3]

**QUESTION 4 (12 marks) START A NEW PAGE**

(a) Find i)  $\int_2^4 \left( \frac{x^3 - 1}{x^2} \right) dx$  [3]

ii)  $\int \sqrt{2x + 3} \, dx$  [3]

(b) Sketch the continuous curve  $y = f(x)$  having all the following properties [3]

$$f(-3) = 12, \quad f(3) = -6, \quad f(0) = 6$$

and  $f'(-3) = f'(3) = 0$

and  $f'(x) < 0$  for  $-3 < x < 3$ ,  $f'(x) > 0$  for  $x > 3$  or  $x < -3$

and  $f''(x) < 0$  for  $x < 0$ ,  $f''(x) > 0$  for  $x > 0$

(c) i) Express  $x^2 + 4x - 12$  in the form  $(x + B)^2 + C$  [1]

1. Hence or otherwise, state the minimum value of  $x^2 + 4x - 12$  and where it occurs [2]

2.

**Question 5 (12 marks)    START A NEW PAGE**

- (a) Given the parabola  $y = 2x - x^2$  and the straight line  $y = 2 - x$ , find
- i) The coordinates of the points of intersection of the parabola and the line [2]
  - ii) The area enclosed between the parabola and the line. [3]
- (b) Let A and B be the fixed points  $(-1,0)$  and  $(2,0)$  and let P be the variable point  $(x,y)$ .
- i) Write down expressions for PA and PB in terms of  $x$  and  $y$  [2]
  - ii) Suppose that P moves so that  $PA=2PB$ . [3]  
Prove that P moves on a circle.
  - iii) Find the centre and radius of this circle [2]

**Question 6 (12 marks)    START A NEW PAGE**

- (a) Find the area bounded by the graph of  $f(x) = x(x - 2)(x + 1)$  and the  $x$ -axis [4]
- (b) Prove that the line  $x + 2y - 4 = 0$  is a tangent to the circle  $(x - 4)^2 + (y - 5)^2 = 20$  [4]
- (c) Find the area bounded by the curve  $y = x^3 + 1$ , the  $y$ -axis and the lines  $y = 1$  and  $y = 9$  [4]

**Question 7 ( 12 Marks)    START A NEW PAGE**

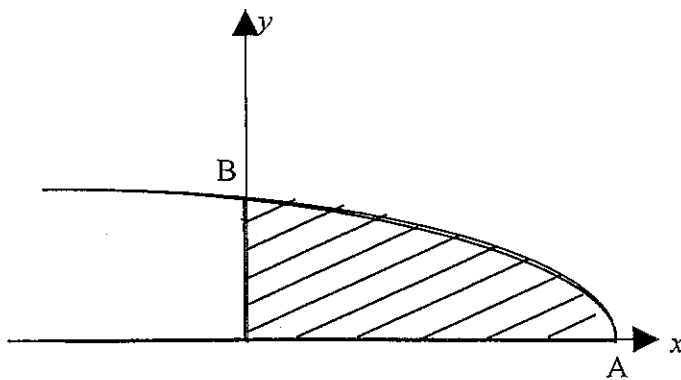
- (a) i) Copy and complete the table of values below for the function  $y = \frac{1}{x}$  [1]

$$y = \frac{1}{x}$$

|        |   |     |   |     |   |
|--------|---|-----|---|-----|---|
| $x$    | 1 | 1.5 | 2 | 2.5 | 3 |
| $f(x)$ |   |     |   |     |   |

- ii) Using the five function values in the table and the Trapezoidal Rule,  
find an approximate value for  $\int_1^3 \frac{1}{x} dx$ , correct to 2 decimal places [3]

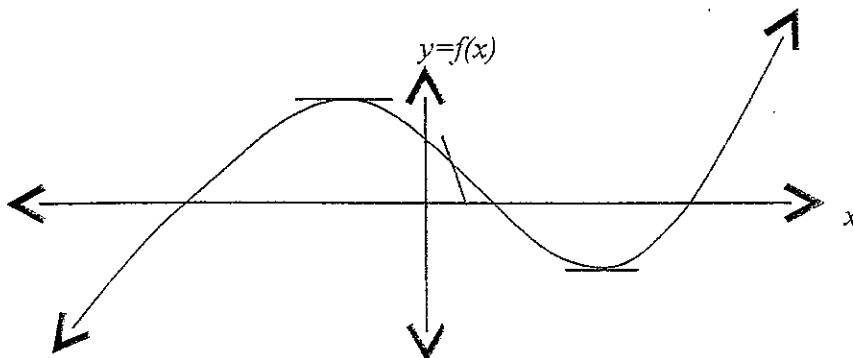
- (b) This sketch represents the curve  $y = \sqrt{9-x}$



Write down the coordinates of A and B,  
the points where the curve cuts the  $x$  and  $y$  axes [2]

Find the volume of the solid formed when the shaded region  
is rotated about the  $x$  axis [4]

- (c) Sketch the gradient function,  $y = f'(x)$ , for the given function  $y = f(x)$ . [2]



# SOLUTIONS TO 2006 MID YEAR

## ASC MATHEMATICS

Q1(a)

RULE

(i)  $y = 3x^2 - \frac{1}{x}$

$y = 3x^2 - x^{-1}$

•  $f(x) = ax^n$

$\frac{dy}{dx} = 6x + x^{-2}$

•  $f'(x) = ax^{n-1}$

$= 6x + \frac{1}{x^2}$  ②

(ii)  $f(x) = (2-4x)^{\frac{1}{2}}$

$f'(x) = \frac{1}{2}(2-4x)^{-\frac{1}{2}}(-4)$

• function of a function

$= -2(2-4x)^{-\frac{1}{2}}$  ②

• chain rule

(iii)  $y = \frac{1-x}{x^2+1}$

$\frac{dy}{dx} = \frac{(x^2+1)(-1) - (1-x)(2x)}{(x^2+1)^2}$

$= \frac{-x^2-1-2x+2x^2}{(x^2+1)^2}$

• Quotient rule

$= \frac{x^2-2x-1}{(x^2+1)^2}$  ②

$y = \frac{u}{v}$

$\frac{dy}{dx} = \frac{v u' - u v'}{v^2}$

$$10) \frac{dy}{dx} = 12x^2 + 1$$

$$\therefore y = \int (12x^2 + 1) dx \quad \cdot \text{indefinite integral}$$

$$y = \frac{12x^3}{3} + x + C$$

$$y = 4x^3 + x + C$$

$$\text{but } y = 10 \text{ when } x = 1$$

$$\therefore 10 = 4(1)^3 + 1 + C \quad \cdot \text{to find } C$$

$$10 = 4 + 1 + C$$

$$C = 5$$

$$\therefore y = 4x^3 + x + 5$$

$$\text{let } x = 2$$

$$y = 4(2)^3 + 2 + 5$$

$$y = 4 \times 8 + 2 + 5$$

$$y = 32 + 7$$

$$y = 39$$

(3)

$$(c) \int_m^{m+2} (2x) dx = 16 \quad \cdot \text{definite integral}$$

$$\left[ x^2 \right]_m^{m+2} = 16$$

$$(m+2)^2 - m^2 = 16$$

$$m^2 + 4m + 4 - m^2 = 16$$

$$4m + 4 = 16$$

$$4m = 12$$

$$\therefore m = 3 \quad (3)$$

Question 2:

(a)  $y = 2x^3 - 3x^2 - 36x + 2$

(i) Stationary points occur when  $\frac{dy}{dx} = 0$

$$6x^2 - 6x - 36 = 0$$

$$6(x^2 - x - 6) = 0$$

$$(x-3)(x+2) = 0$$

$$\therefore x = 3, -2$$

When  $x = -2$

$$\frac{d^2y}{dx^2} = 12x - 6$$

$$= 12(-2) - 6$$

$$= -30$$

$< 0$  concave down

$$y = 2(-8) - 3(4) - 36(-2) + 2$$
$$= 46$$

$\therefore$  At  $(-2, 46)$  a relative maximum stationary point occurs.

when  $x = 3$

$$\frac{d^2y}{dx^2} = 12x - 6$$

$$= 12(3) - 6$$

$$= 30$$

$> 0$  concave up.

$$y = 2(27) - 27 - 36 \times 3 + 2$$
$$= -79$$

$\therefore$  At  $(3, -79)$  a relative minimum stationary point occurs

1 mark  $x = -2, 3 \in \mathbb{R}$   $\frac{dy}{dx} = 6x^2 - 6x - 36$ ,  $\frac{d^2y}{dx^2} = 12x - 6$

1 mark  $(-2, 46)$  rel min

1 mark  $(3, -79)$  rel max

Section very well done.  
Careless errors cost marks  
especially omitting  $(-, -)$ .

(a) (ii) Possible points of inflexion occur when  $\frac{d^2y}{dx^2} = 0$

$$\frac{d^2y}{dx^2} = 12x - 6$$

$$0 = 6(2x - 1)$$

$$x = \frac{1}{2}$$

$$\left. \begin{array}{l} \text{when } x < \frac{1}{2} \quad \frac{d^2y}{dx^2} < 0 \quad (\text{TEST } x=0) \\ x > \frac{1}{2} \quad \frac{d^2y}{dx^2} > 0 \quad (\text{TEST } x=1) \end{array} \right\} \text{1 mark}$$

Concavity changes either side of  $x = \frac{1}{2}$ .

When  $x = \frac{1}{2}$

$$y = 2x^3 - 3x^2 - 36x + 2$$

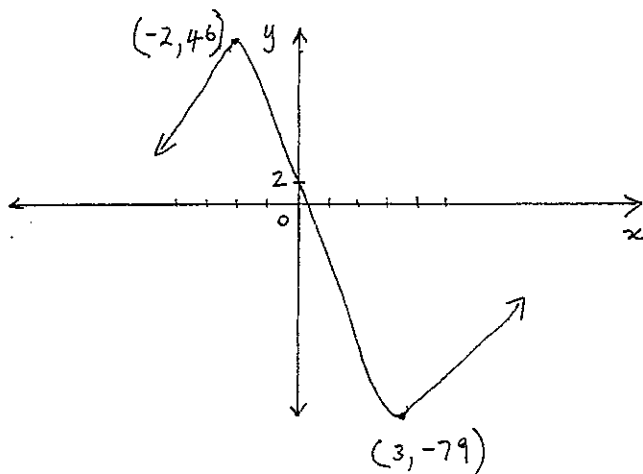
$$= \frac{2}{8} - \frac{3}{4} - 18 + 2$$

$$= -16\frac{1}{2}$$

Many students did not indicate the purpose of the 2<sup>ND</sup> derivative test  
You WILL lose a mark  
NEXT TIME

$\therefore$  Point of inflexion occurs at  $(\frac{1}{2}, -16\frac{1}{2})$  ← 1 mark

(iii)



1 mark  
y-intercept clearly shown

1 mark  
basic sketch of a cubic fn with max, min and axes labelled

(iv) The function is decreasing in the domain  $-2 < x < 3$  1 mark

some students misread the question.

A fn is decreasing where  $\frac{dy}{dx} < 0$

$$(b) \quad 2x^2 - 8x - 25 = 0$$

If  $\alpha, \beta$  are the roots then

$$(x - \alpha)(x - \beta) = 0$$

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0$$

$$\text{cf} \quad x^2 - \frac{8}{2}x - \frac{25}{2} = 0$$

$$x^2 - 4x - \frac{25}{2} = 0$$

$$(i) \quad \alpha + \beta = -(-4) \\ = 4$$

\_\_\_\_\_ 1 mark

$$(ii) \quad \alpha\beta = -\frac{25}{2} \quad (-12\frac{1}{2})$$

\_\_\_\_\_ 1 mark

$$(iii) \quad \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$$

\_\_\_\_\_ 1 mark

$$= 4^2 - 2(-\frac{25}{2})$$

$$= 41$$

\_\_\_\_\_ 1 mark

IN General, question well done.

Students lost marks in part (iii) if they demonstrated poor understanding. eg, a real number squared is always positive.

The sum of two positive numbers can only be positive.

A significant number of students obtained a negative solution.  
IMPOSSIBLE for real numbers.

No marks awarded when solution defies logic.

### Question 3. (12 marks).

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(a) (i)  $V = x^2 y = 32$   
 $\therefore y = \frac{32}{x^2}$  (1 mark)

(ii)  $A = x^2 + 4xy$   
 $= x^2 + 4x \left( \frac{32}{x^2} \right)$  (1)  
 $= x^2 + \frac{128}{x}$

(iii)  $A = x^2 + 128x^{-1}$

$\frac{dA}{dx} = 2x - 128x^{-2}$

$\frac{d^2A}{dx^2} = 2x + 256x^{-3}$

For max/min value  $\frac{dA}{dx} = 0$  (1)

$\therefore \frac{2x - 128}{x^2} = 0$

$\therefore x = 4$

If  $x = 4$ ,  $\frac{d^2A}{dx^2} > 0$  (1)

$\therefore$  Dimensions are 4cm  $\times$  4cm  $\times$  2cm.

(b) (i)  $x^2 = 4(2-y)$

$x^2 = 8 - 4y$

$x^2 = -4y + 8$

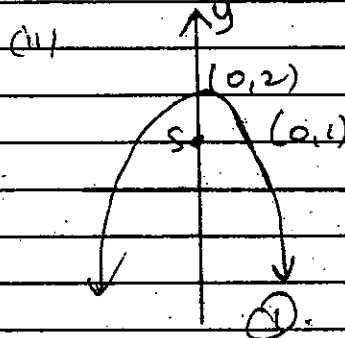
$x^2 = -4(y-2)$  (1)

$\therefore$  Vertex is (0,2)

$\therefore a = -1$

$\therefore a = 1$

$\therefore$  Focus is (0,1) (1)



(ii)  $x^2 = -4y + 8$

$4y = -x^2 + 8$

$y = -\frac{x^2}{4} + 2$

$\frac{dy}{dx} = -\frac{x}{2}$  (1)

At  $x=2$ ,  $\frac{dy}{dx} = -1$  (m)

$m_1 \cdot m_2 = 1$  (1)

Equation of normal

is  $y-1 = 1(x-2)$

$y = x-1$  (1)

For Areas, etc

You MUST write

$\frac{dA}{dx} = 0$  or  $A'$

$A'$  is NOT

acceptable

Must write the

Statement

you MUST test

the value

It's Volume, so

you must write

3 dimensions. (1)

NOT area.

This question is

to confuse. You

need to factorise

negative out.

The graph should

represent your

answers in (a).

You must make

y the subject &

differentiate

# Question 4

12

$$\begin{aligned} \text{a) (i)} \int_2^4 \left( \frac{x^3-1}{x^2} \right) dx &= \int_2^4 \left( \frac{x^3}{x^2} - \frac{1}{x^2} \right) dx \\ &= \int_2^4 (x - x^{-2}) dx \\ &= \left[ \frac{x^2}{2} - \frac{x^{-1}}{-1} \right]_2^4 \\ &= \left[ \frac{x^2}{2} + \frac{1}{x} \right]_2^4 \\ &= \left( \frac{16}{2} + \frac{1}{4} \right) - \left( \frac{4}{2} + \frac{1}{2} \right) \\ &= 8\frac{1}{4} - 2\frac{1}{2} \\ &= 5\frac{3}{4} \end{aligned}$$

① simplifying the expression to be integrated

① integrating ③

① sub. and simplify

Note: \* You cannot integrate a product (or a quotient at the moment)

\* Many students need to take care with notation.

$$\text{(ii)} \int \sqrt{2x+3} \, dx = \int (2x+3)^{\frac{1}{2}} dx$$

$$\begin{aligned} &= \frac{(2x+3)^{\frac{3}{2}}}{\frac{3}{2} \times 2} + C \quad \text{①} \\ &= \frac{\sqrt{(2x+3)^3}}{3} + C \quad \text{① integrating} \end{aligned}$$

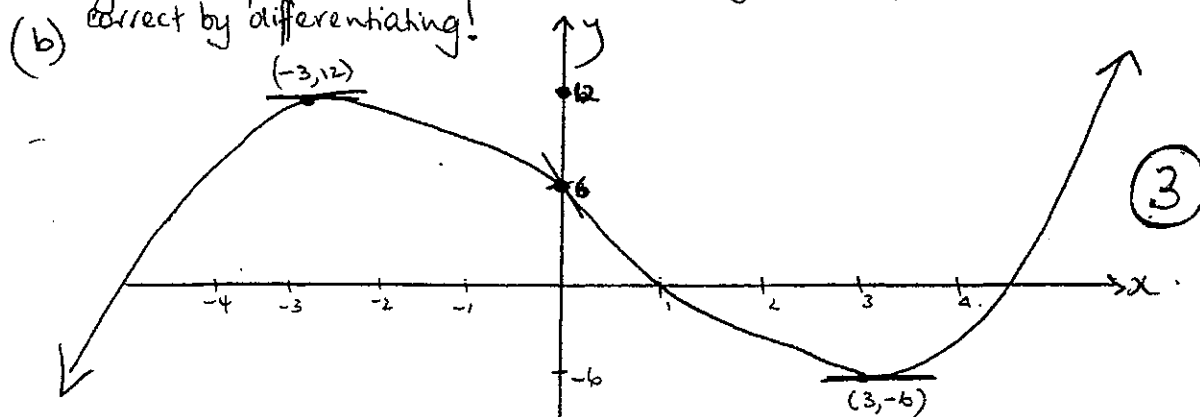
Notes:

\* Indefinite integrals (with no limits) require a constant of integration, 'C'.

\* Always check primitives are correct by differentiating!

① simplifying / correct form!

③



① for shape (max + min)

① for showing clearly point of inflexion at (0, 6)

① smooth curve / scale

Q4 continued.

c) (i)  $x^2 + 4x - 12 = (x^2 + 4x + 4) - 12 - 4$   
 $= (x+2)^2 - 16$  ①

Note:  
\*The 'hence'  
in part (ii)  
means use  
(i) to solve  
part (ii)!

(ii)  $\therefore$  minimum value is  $-16$  when  $x = -2$  ①

OR  $x = -\frac{b}{2a}$ ,  $b=4$ ,  $a=1$

$\therefore x = -\frac{4}{2}$

$= -2$

i.e. vertex occurs at  $x = -2$  on

$f(-2) = (-2)^2 + 4(-2) - 12$

$= 4 - 8 - 12$

$= -16$

$\therefore$  minimum of  $-16$  at  $x = -2$  ①

Note:  $(x+2)^2 \geq 0$  for all  $x$

$\therefore (x+2)^2 - 16$  is at  
least  $-16$  (when  $x = -2$ )

parabola  $y = x^2 + 4x - 12$

(where  $a=1$  i.e. has  
minimum turning point) ①

OR Let  $y = x^2 + 4x - 12$

For min. value,  $\frac{dy}{dx} = 0$ ,  $\frac{d^2y}{dx^2} > 0$

$\frac{dy}{dx} = 2x + 4$

i.e.  $0 = 2x + 4$

$\therefore x = -2$

At  $x = -2$

$y = (-2)^2 + 4(-2) - 12$

$= -16$

$\therefore$  minimum of  $-16$  at  $x = -2$ ! ①

$f(x)$  value

For  
①  
mark?

Note: \* Stating the coordinates of the minimum  
value is not enough for full marks.

\* You had to interpret the coordinates  
 $\rightarrow$  know that the minimum is the y-value!

Solutions for Mathematics Half Yearly 2006 Year 12

**Question 5**

a) i)  $y = 2x - x^2$ ,  $y = 2 - x$

equating the two equations

$$2x - x^2 = 2 - x$$

$$x^2 - 3x + 2 = 0$$

$$(x-2)(x-1) = 0$$

$$x = 1, 2$$

1 mark

∴ points of intersection are (1,1) and (2,0)

1 mark

- The question asked for the coordinates of the points of intersection, so you needed to write answers as ordered pairs of numbers

ii)

Write Area=

$$\text{Area} = \int_1^2 [(2x - x^2) - (2 - x)] dx$$

1 mark

$$= \int_1^2 (3x - x^2 - 2) dx$$

$$= \left[ \frac{3x^2}{2} - \frac{x^3}{3} - 2x \right]_1^2$$

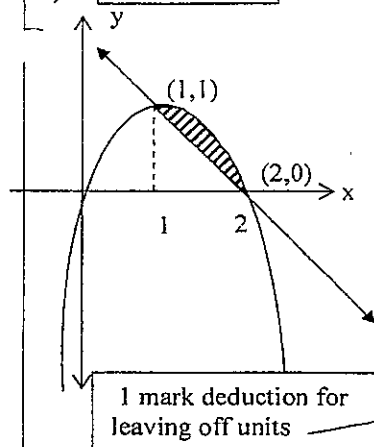
1 mark

$$= \left( \frac{12}{2} - \frac{8}{3} - 4 \right) - \left( \frac{3}{2} - \frac{1}{3} - 2 \right)$$

$$= \frac{-2}{3} - \frac{5}{6}$$

$$= \frac{1}{6} \text{ units}^2$$

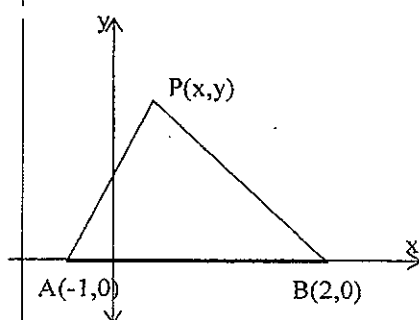
1 mark



Always simplify the expression before integrating by collecting like terms

- You must write the subject of the equation. AREA=
- There is generally a poor use of notation. Once you have integrated,  $\int$  and the  $dx$  are no longer written
- Many students use absolute value signs incorrectly. They should appear at the beginning and not taken off until the very last step

b)



i)  $PA = \sqrt{(x+1)^2 + y^2}$

1 mark each

$PB = \sqrt{(x-2)^2 + y^2}$

ii)  $PA = 2PB$

$$\sqrt{(x+1)^2 + y^2} = 2\sqrt{(x-2)^2 + y^2}$$

1 mark

$$x^2 + 2x + 1 + y^2 = 4x^2 - 16x + 16 + 4y^2$$

$$3x^2 - 18x + 3y^2 + 15 = 0$$

$$x^2 - 6x + y^2 + 5 = 0$$

1 mark

$$(x^2 - 6x + 9) + y^2 + 5 + 9 = 0$$

$$(x-3)^2 + y^2 = 4$$

1 mark

which is a circle of the form

$$(x-h)^2 + (y-k)^2 = r^2$$

iii) Centre (3,0)

Radius= 2 units

1 mark each

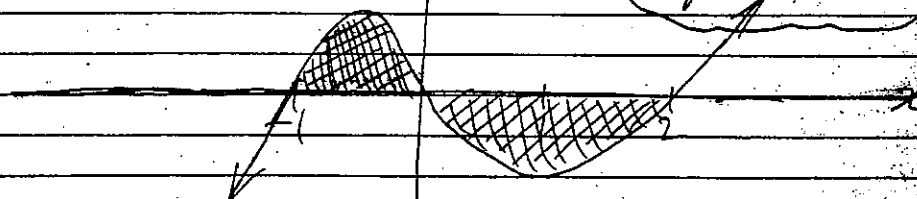
- Read the question carefully. You were asked to find PA and PB NOT  $PA^2$  and  $PB^2$ . You lost a mark for going on and doing some strange simplification as you cannot have 2 answers for the one question

- Many students forgot to square the 2 when squaring both sides

- To show it is a circle you should compare your answer to the general form. Not just say it is a circle

Q6.  $f(x) = x(x-2)(x+1) = x(x^2 - x - 2)$   
 (a) let  $f(x) = 0$   $= x^3 - x^2 - 2x$   
 $0 = x(x-2)(x+1)$   
 $x = 0, 2, -1$

• always draw a quick sketch



$$A = \int_{-1}^0 (x^3 - x^2 - 2x) dx + \left| \int_0^2 (x^3 - x^2 - 2x) dx \right|$$

$$A = \left[ \frac{x^4}{4} - \frac{x^3}{3} - x^2 \right]_{-1}^0 + \left| \left[ \frac{x^4}{4} - \frac{x^3}{3} - x^2 \right]_0^2 \right|$$

$$A = (0 - 0 - 0) - \left( \frac{1}{4} - \frac{1}{3} - 1 \right) + \left| \left( 4 - \frac{8}{3} - 4 \right) \right| = 0$$

$$A = -\frac{5}{12} + \left| -2\frac{2}{3} \right|$$

$$A = -\frac{5}{12} + 2\frac{2}{3}$$

$$A = 3\frac{1}{12} \text{ units}^2$$

• always take the absolute value because we can't have a negative area.

④

$$60) (x-4)^2 + (y-5)^2 = 20$$

centre  $(4, 5)$ , radius  $= \sqrt{20}$

$$r = 2\sqrt{5} \text{ units}$$

Find the perp. dist. from  $(4, 5)$  to the line  $x + 2y - 4 = 0$

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

the perp. distance formula

$$d = \frac{|4 + 10 - 4|}{\sqrt{1^2 + 2^2}}$$

$$d = \frac{10}{\sqrt{5}} \text{ units}$$

$$d = \frac{10}{\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}}$$

$$d = \frac{10\sqrt{5}}{5}$$

$$d = 2\sqrt{5} \text{ units}$$

④

Now,  $d = 2\sqrt{5}$  and the radius is  $2\sqrt{5}$  so, the line  $x + 2y - 4 = 0$  must be a tangent to the circle.

• you must write your conclusion i.e. the reason why the line is a tangent

$$6(b) (x-4)^2 + (y-5)^2 = 20 \quad \text{--- ①}$$

$$x + 2y - 4 = 0 \quad \text{--- ②}$$

• simultaneous equations

From ②  $x = 4 - 2y$

Subst. from ② into ①

$$(4 - 2y - 4)^2 + (y - 5)^2 = 20$$

$$(2y)^2 + y^2 - 10y + 25 = 20$$

$$4y^2 + y^2 - 10y + 25 = 20$$

$$5y^2 - 10y + 5 = 0$$

$$y^2 - 2y + 1 = 0$$

$$(y - 1)(y - 1) = 0$$

$$y = 1, y = 1$$

• must write your conclusion

Subst. into  $x = 4 - 2y$

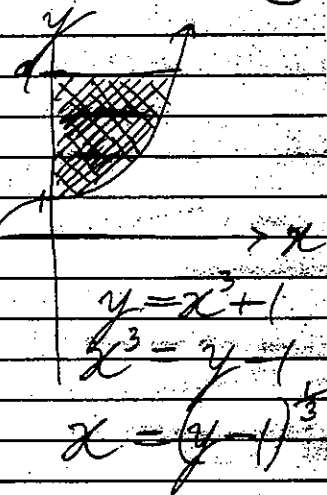
$$\text{let } y = 1, x = 4 - 2$$

$$x = 2$$

(2, 1) is the only point of intersection  
so the line  $x + 2y - 4 = 0$  is a  
tangent to the circle. (4)

$$6(c) A = \int_0^8 x \, dy$$

• draw a diagram



$$A = \int_1^9 (y-1)^{1/3} \, dy$$

$$A = \left[ \frac{3}{4} (y-1)^{4/3} \right]_1^9$$

$$A = \left[ \frac{3}{4} (8)^{4/3} - 0 \right]$$

$$A = \frac{3}{4} (16)$$

$$A = 12 \text{ units}^2 \quad (4)$$

### Question 7

a)

| x    | 1 | 1.5                  | 2                    | 2.5                  | 3                    |
|------|---|----------------------|----------------------|----------------------|----------------------|
| f(x) | 1 | $\frac{2}{3}$ or 0.6 | $\frac{1}{2}$ or 0.5 | $\frac{2}{5}$ or 0.4 | $\frac{1}{3}$ or 0.3 |

b) For one application trapezoidal rule says

$$\text{Area} \approx \frac{h}{2}(f(a) + f(b))$$

1 mark

Here there are 4 applications so  $h = \frac{3-1}{4} = \frac{1}{2}$

1 mark

$$\begin{aligned} \therefore \int_1^3 \frac{1}{x} dx &\approx \frac{1}{2} \left( 1 + \frac{2}{3} \right) + \frac{1}{4} \left( \frac{2}{3} + \frac{1}{2} \right) + \frac{1}{4} \left( \frac{1}{2} + \frac{2}{5} \right) + \frac{1}{4} \left( \frac{2}{5} + \frac{1}{3} \right) \\ &\approx \frac{1}{4} \times 4 \times \frac{7}{15} \\ &\approx 1.11666..... \\ &= 1.12 \text{ (to 2 d.p.)} \end{aligned}$$

1 mark

1 mark

OR in table form

| x              | $f(x) = \frac{1}{x}$ | Weighting factor | WF $\times$ f(x) |
|----------------|----------------------|------------------|------------------|
| 1              | 1                    | 1                | 1                |
| $1\frac{1}{2}$ | $\frac{2}{3}$        | 2                | $\frac{4}{3}$    |
| 2              | $\frac{1}{2}$        | 2                | 1                |
| $2\frac{1}{2}$ | $\frac{2}{5}$        | 2                | $\frac{4}{5}$    |
| 3              | $\frac{1}{3}$        | 1                | $\frac{1}{3}$    |

4 applications

width of each application =  $\frac{1}{2}$

1 mark

1 mark

$$\begin{aligned} \int_1^3 \frac{1}{x} dx &\approx \frac{1}{2} \times \text{width of application} \times \text{sum} \\ &\approx \frac{1}{4} \times 4 \times \frac{7}{15} \\ &\approx 1.11666..... \\ &= 1.12 \text{ (to 2 d.p.)} \end{aligned}$$

$$\text{Sum} = 4 \frac{7}{15}$$

1 mark

1 mark

• You should be working with exact values until the last step as you create rounding errors as you work through the question

• When you have to complete a table, you actually need to find the values, by simplifying your substitutions

• This was done poorly as many students learn a long formula but don't actually recognize how many applications of the rule they are doing. You should show the number of applications

• Here there are 4 applications and each one has a width of  $\frac{1}{2}$ . So  $h = \frac{1}{2}$

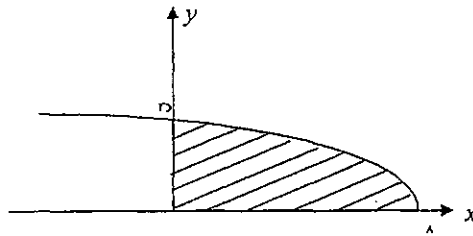
• This is not an area question and so there should be no units on the answer

• The subject of the equations should be  $\therefore \int_1^3 \frac{1}{x} dx \approx$  and as it is an approximate method you must use approximately equal to signs NOT =

• There was a mark penalty for incorrect use of units and approx equal signs

• Be careful not to confuse Simpson's rule and the trapezoidal rule

b)  $y = \sqrt{9-x}$



i) When  $x=0$ ,  $y=3$   
So B is (0,3)

1 mark

$$0 = \sqrt{9-x}$$

When  $y=0$ ,  $0^2 = 9-x$

$$x = 9$$

So A is (9,0)

1 mark

$$\pi \int_0^9 (\sqrt{9-x})^2 dx$$

1 mark for setting up the volume question

$$= \pi \int_0^9 (9-x) dx$$

ii) Volume =  $\pi \left[ 9x - \frac{x^2}{2} \right]_0^9$

1 mark for correct integration

$$= \pi \left[ 81 - \frac{81}{2} - 0 \right]$$

$$= \frac{81\pi}{2} \text{ units}^3$$

1 mark for units

1 mark for answer. It is best left in exact form ie. In terms of pi

- The question required coordinates. You received 1 mark only for  $x=9$  and  $y=3$

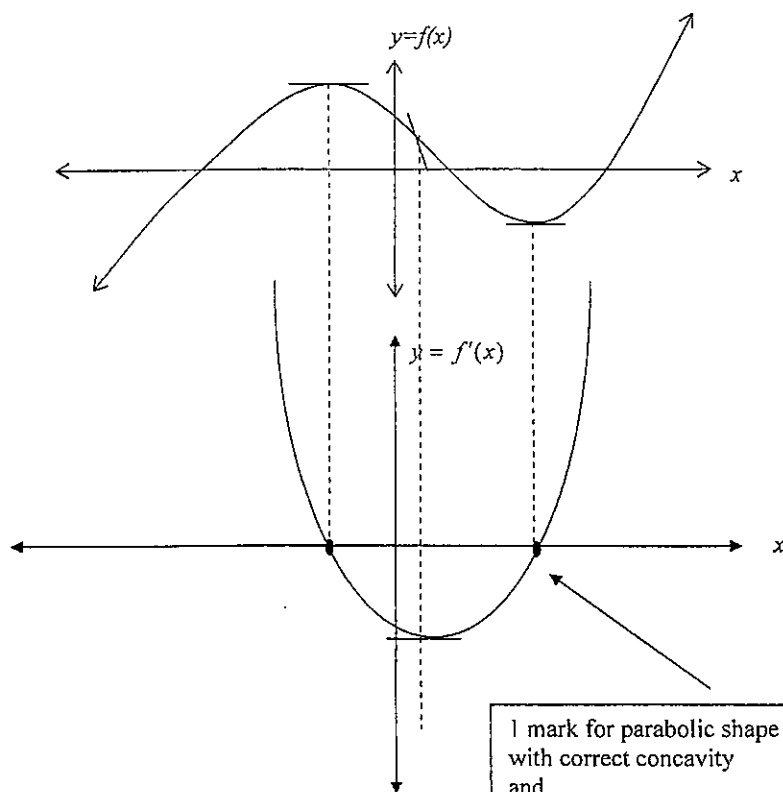
- This was a volume around the x-axis, but many students did not read the question and went around the y-axis. Many also found area instead of volume

- Again you need a subject for your equation  
Volume =

- Don't forget units cubed for volume

- Many students forgot to square the function and so created problems for the integration

Question 7 c)



1 mark for parabolic shape  
with correct concavity  
and  
1 mark for roots and vertex  
in correct position

- When drawing a derivative function, you must redraw the original function and clearly show where the critical points line up
- Most students didn't bother drawing the first function and so as a marker I can't check whether or not you have your roots and turning point in the right place.
- PLEASE use a ruler when drawing diagrams and make them big